Representation Regularization and Theoretical Mechanics of Joint-Embedding Predictive Architectures for Multi-Modal Brain MRI: A Comprehensive Analysis of Label Efficiency, Non-Collapse Dynamics, and Out-of-Distribution Generalization

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Abstract

Joint-Embedding Predictive Architectures (JEPA) represent a paradigm shift in self-supervised visual representation learning, operating entirely in latent embedding space without pixel-level image reconstruction. Standard I-JEPA prevents latent representation collapse through an Exponential Moving Average (EMA) momentum update of a teacher target encoder. Recent theoretical formulations—specifically SIGReg (Sketched Isotropic Gaussian Regularization in LeJEPA) and VISReg (Variance-Invariance-Sketching Regularization)—introduce analytical feature regularization penalties that enable *heuristic-free* single-encoder JEPA training without momentum teachers. In this paper, we present a mathematically rigorous and comprehensive evaluation of I-JEPA, SigReg JEPA, VisReg JEPA, a 2D Residual UNet baseline, and a state-of-the-art 2D nnU-Net baseline with deep supervision on 2D multi-modal Glioma MRI (`BraTS GLI`). We detail the mathematical formulations and architectural micro-designs across all models, including internal residual skip connections, multi-scale encoder-to-decoder concatenation skip connections, pure bottleneck transposed convolutional decoding, stochastic patch block masking with batch-uniform collation, Random Modality Dropout with guaranteed active channel fallbacks, hyperspherical projection sketching, and gradient-clipped optimization schedules. We evaluate latent collapse metrics (Effective Feature Rank Rank_{eff}, Average Pairwise Cosine Similarity S_{cosine}), downstream segmentation metrics (Dice Similarity Coefficient, Intersection over Union, 95th Percentile Hausdorff Distance HD95), and out-of-distribution transfer across three clinical regimes: (1) Full-Data Benchmark, (2) Low-Data Label Efficiency (1% to 100% labels), and (3) Out-of-Distribution (OOD) Scanner Shift and Zero-Shot Cross-Pathology Transfer on Meningioma MRI (`BraTS-MEN-RT`). Our empirical findings establish that under extreme label scarcity (1% labels / 63 training slices), SigReg JEPA achieves a top downstream Test Dice of 0.5933, outperforming fully supervised nnU-Net (0.3847) by $+20.86\%$ absolute Dice ($+54.2\%$ relative improvement) while executing fine-tuning $1.9\times$ faster per epoch (0.75 s/epoch vs 1.42 s/epoch). Furthermore, under zero-padded missing modality shifts, self-supervised JEPAs preserve zero-shot transferability (0.0700–0.0929 Dice) whereas standard UNet completely collapses (0.0151 Dice), proving that analytical variance-covariance regularization builds

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modality-invariant, high-rank representations while eliminating momentum teacher overhead.

1 Introduction and Theoretical Framework

Automated segmentation of brain gliomas from multi-modal Magnetic Resonance Imaging (MRI)—comprising T1-weighted (T1), post-contrast T1-weighted (T1c), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR) sequences—is essential for clinical neuro-oncology diagnosis, surgical boundary planning, and longitudinal monitoring [Baid et al., 2024]. Supervised deep convolutional architectures, such as Residual UNet [Ronneberger et al., 2015] and nnU-Net [Isensee et al., 2021], achieve high accuracy when extensive pixel-wise ground-truth annotations are available. However, expert manual voxel delineation is labor-intensive, scarce, and susceptible to inter-observer variability.

Self-Supervised Learning (SSL) addresses this bottleneck by pre-training representation encoders on unannotated MRI volumes prior to downstream task fine-tuning. Unlike Masked Autoencoders (MAE) [He et al., 2022] that reconstruct high-frequency pixel intensities in pixel space, Joint-Embedding Predictive Architectures (JEPA) [Assran et al., 2023] operate exclusively in latent embedding space. JEPA models predict target patch representations from context patch representations, forcing the encoder to learn abstract semantic structures rather than low-level noise.

A central challenge in energy-based self-supervised architectures is preventing *latent representation collapse*—a trivial state where the encoder maps all input patches to a constant vector $z = c \in \mathbb{R}^D$, yielding zero prediction error $\mathcal{L} = 0$ without learning meaningful semantics.

Standard I-JEPA [Assran et al., 2023] prevents collapse through an Exponential Moving Average (EMA) momentum update of a target teacher encoder $E_{\bar{\theta}}$. However, EMA updating is a heuristic mechanism that introduces computational memory overhead, requires hyperparameter tuning of the momentum parameter m, and lacks formal non-collapse guarantees.

Recent theoretical formulations derived from statistical distribution matching and sketching principles, specifically SIGReg [Balestriero and LeCun, 2025] and VISReg [Wu et al., 2026], introduce analytical feature regularization penalties applied directly to online encoder activations. By constraining representations to match isotropic Gaussian distributions via 1D random projections, these methods establish a *heuristic-free* single-encoder JEPA framework that eliminates EMA target teachers.

In this paper, we provide a complete mathematical and empirical breakdown of JEPA architectures for multi-modal brain MRI segmentation. We formalize loss objectives, backpropagation decoupling, training protocols, probing metrics, and evaluation procedures across full-data, low-data, and out-of-distribution transfer settings.

2 Mathematical Methodology and Architecture Formulations

2.1 Multi-Modal Input Processing and Random Modality Dropout

Let $X \in \mathbb{R}^{B\times 4\times H\times W}$ represent a batch of 2D axial MRI slices with spatial dimensions $H = W = 240$ and $C = 4$ co-registered sequence channels: $X = [X_{\mathrm{T1}}, X_{\mathrm{T1c}}, X_{\mathrm{T2}}, X_{\mathrm{FLAIR}}]$.

Physical Sequence Characteristics & Clinical Tumor Pathophysiology Glioma diagnosis and boundary delineation require multi-parametric MRI because different pulse sequences probe orthogonal biological tissue properties [Baid et al., 2024]:

- **T1-weighted (T1)**: Characterized by short repetition time (TR) and echo time (TE), T1 reflects longitudinal spin-lattice relaxation (T_1). It yields sharp contrast between healthy gray matter, white matter, and cerebrospinal fluid (CSF), providing anatomical reference geometry.
- **Contrast-Enhanced T1-weighted (T1c)**: Administered via intravenous Gadolinium-based contrast agents (e.g., Gd-DTPA). Paramagnetic gadolinium ions accelerate local water proton T_1 relaxation, producing intense signal enhancement strictly where the blood-brain

barrier (BBB) is structurally disrupted by tumor neovascularization. T1c delineates the actively proliferating, viable angiogenic tumor boundary.

- **T2-weighted (T2)**: Utilizing long TR and TE, T2 probes transverse spin-spin relaxation (T_2). It is highly sensitive to free and bound water accumulation, clearly depicting diffuse hyperintense vasogenic edema and bulk mass effect.
- **Fluid-Attenuated Inversion Recovery (FLAIR)**: Applies an initial $180°$ inversion pulse with an inversion time $TI \approx 2000\,\text{ms}$ precisely timed to the null point of free water. By suppressing the bright signal of ventricular CSF, FLAIR isolates non-enhancing peritumoral infiltrative edema and satellite tumor margins that would otherwise be masked by adjacent ventricles.

Prior to slice extraction, non-zero brain tissue voxels are Z-score normalized independently per 3D volume sequence:

$$\hat{X}_c(v) = \frac{X_c(v) - \mu_c}{\sigma_c + \epsilon}, \qquad \forall c \in \{\mathrm{T1}, \mathrm{T1c}, \mathrm{T2}, \mathrm{FLAIR}\} \tag{1}$$

where μ_c and σ_c denote the empirical mean and standard deviation computed strictly over non-zero intracranial voxels in volume sequence c, and $\epsilon = 10^{-8}$ ensures numerical stability.

Random Modality Dropout (RMD) with Guaranteed Non-Empty Fallback In routine clinical oncology, complete 4-sequence acquisitions are frequently compromised by emergency patient agitation, metallic implants, acquisition time constraints, or severe renal impairment contraindicating gadolinium injection (nephrogenic systemic fibrosis) [Havaei et al., 2017, Dorent et al., 2019]. Models naively trained on complete 4-channel tensors develop brittle inter-channel co-dependencies and collapse when modalities are omitted.

To instill intrinsic modality invariance during training, we propose **Random Modality Dropout (RMD)** with a guaranteed non-empty fallback mechanism, formalized in Algorithm 1.

Algorithm 1 Random Modality Dropout with Guaranteed Non-Empty Fallback

Require: Normalized multi-channel slice batch $\hat{X} \in \mathbb{R}^{B\times C\times H\times W}$, dropout probability $p_{\text{drop}} \in [0, 1)$, training flag $\tau \in \{0, 1\}$.
Ensure: Modality-augmented batch $\tilde{X} \in \mathbb{R}^{B\times C\times H\times W}$.

if $\tau = 0$ **or** $p_{\text{drop}} \leq 0$ **then**
  return $\hat{X}$ ▷ Deactivated during inference and validation
end if
Sample channel mask: $\mathbf{M} \leftarrow \mathbb{I}\left(\mathbf{U} > p_{\text{drop}}\right)$ where $\mathbf{U} \sim \mathcal{U}(0, 1)^{B\times C\times 1\times 1}$
Identify completely zeroed slices: $\mathbf{A}_{\text{zero}} \leftarrow \left(\sum_{c=1}^{C} \mathbf{M}_{:,c,1,1} == 0\right) \in \{0, 1\}^B$
Sample fallback channels: $c^* \sim \text{Uniform}(\{1, \ldots, C\})^B$
Construct fallback one-hot mask: $\mathbf{M}_{\text{fallback}} \leftarrow \text{OneHot}(c^*) \in \mathbb{R}^{B\times C\times 1\times 1}$
Reconcile mask: $\mathbf{M} \leftarrow \text{where}(\mathbf{A}_{\text{zero}}, \mathbf{M}_{\text{fallback}}, \mathbf{M})$
return $\tilde{X} = \hat{X} \odot \mathbf{M}$ ▷ Element-wise broadcast multiplication

Spatial Data Augmentation Randomization During downstream fine-tuning, training slices undergo stochastic spatial transformations:

- **Random Horizontal Flip**: Spatial axis 0 inverted with probability $p = 0.5$.
- **Random Vertical Flip**: Spatial axis 1 inverted with probability $p = 0.5$.
- **Random Affine Rotation**: Uniform spatial rotation angle $\theta \sim \mathcal{U}(-17.2°, +17.2°)$ (range $0.3\,\text{rad}$) with probability $p = 0.5$, applying bilinear interpolation to image intensities and nearest-neighbor interpolation to discrete segmentation ground-truth.

2.2 Vision Transformer Backbone Architecture & Internal Skip Connections

The vision encoder E_θ processes 2D multi-modal slices using non-overlapping $P \times P = 16 \times 16$ patch embeddings. A 2D slice with $H = W = 240$ is partitioned into a spatial grid of patch tokens:

$$N = \left(\frac{H}{P}\right) \times \left(\frac{W}{P}\right) = \left(\frac{240}{16}\right) \times \left(\frac{240}{16}\right) = 15 \times 15 = 225 \text{ patches} \tag{2}$$

A 2D convolution with kernel size 16, stride 16, and 4 input channels maps each $16 \times 16 \times 4$ patch to embedding dimension $D = 384$. Learnable 1D spatial positional embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{1\times 225\times 384}$, initialized from a truncated normal distribution $\mathcal{TN}(0, 0.02^2)$, are added:

$$\mathbf{Z}^{(0)} = \text{Conv2d}_{16\times 16}(\tilde{X}) + \mathbf{E}_{\text{pos}} \in \mathbb{R}^{B\times 225\times 384} \tag{3}$$

The patch tokens pass through $L = 8$ Transformer Encoder blocks. Each block employs a **Pre-Layer Normalization (Pre-LN)** design featuring internal residual skip connections around both the attention and feed-forward sub-layers:

$$\mathbf{Z}'^{(l)} = \mathbf{Z}^{(l-1)} + \text{MHSA}\left(\text{LN}(\mathbf{Z}^{(l-1)})\right) \tag{4}$$

$$\mathbf{Z}^{(l)} = \mathbf{Z}'^{(l)} + \text{MLP}\left(\text{LN}(\mathbf{Z}'^{(l)})\right), \qquad l = 1, \dots, 8 \tag{5}$$

The residual shortcut additions in Equations (4) and (5) preserve identity gradient highways throughout the depth of the encoder, preventing gradient attenuation. The Multi-Head Self-Attention (MHSA) module operates across $h = 6$ attention heads (head dimension $d_k = D/h = 64$):

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{6}$$

The MLP sub-layer consists of a two-layer feed-forward network with expansion ratio 4 ($D_{\text{ff}} = 1536$) and GELU non-linearities: $\text{MLP}(x) = \text{Linear}_{1536\to 384}\left(\text{GELU}\left(\text{Linear}_{384\to 1536}(x)\right)\right)$.

Vectorized Subset Patch Encoding When encoding context patches during self-supervised pre-training, only context indices $\mathcal{I}_{\text{ctx}} \subset \{1, \dots, N\}$ ($|\mathcal{I}_{\text{ctx}}| = N_{\text{ctx}}$) are passed to the encoder. To completely prevent attention leakage across masked target patches without inefficient $O(B)$ Python loops, we vectorize subset selection using `torch.gather`:

$$\mathbf{Z}^{(0)}_{\text{ctx}} = \text{gather}\left(\mathbf{Z}^{(0)},\ \text{dim} = 1,\ \text{index} = \mathcal{I}_{\text{ctx}}\right) \in \mathbb{R}^{B\times N_{\text{ctx}}\times 384} \tag{7}$$

Self-attention is then computed strictly over context tokens, guaranteeing zero computational cost and zero information leakage from masked target regions.

2.3 Stochastic Patch Masking: JEPAMaskingTransform

The self-supervised prediction task requires partitioning the 15×15 grid into context and target patch sets. We implement a stochastic block masking transform (`JEPAMaskingTransform`) designed to satisfy three criteria: (1) realistic spatial context, (2) zero overlap between context and targets, and (3) uniform tensor dimensions for dense batch collation.

1. **Target Block Sampling**: We sample $K = 4$ target blocks. Each target block has a fixed spatial geometry of $H_{\text{tgt}} \times W_{\text{tgt}} = 5 \times 5 = 25$ patches ($N_{\text{tgt}} = 25$). The top-left corner coordinates (t_k, l_k) for each block $k \in \{1, \dots, K\}$ are sampled uniformly:

$$t_k \sim \mathcal{U}\{0,\ H_{\text{grid}} - H_{\text{tgt}}\}, \quad l_k \sim \mathcal{U}\{0,\ W_{\text{grid}} - W_{\text{tgt}}\} \tag{8}$$

   The target patch indices for block k are $\mathcal{I}_{\text{tgt},k} = \{(t_k + r)W_{\text{grid}} + (l_k + c) \mid 0 \le r < 5,\ 0 \le c < 5\}$. The composite target set is $\mathcal{I}_{\text{tgt}} = \bigcup_{k=1}^{K} \mathcal{I}_{\text{tgt},k}$.

2. **Candidate Context Window**: A large candidate context block of spatial dimension $H_{\text{ctx}} \times W_{\text{ctx}} = 14 \times 14 = 196$ patches is sampled from the grid:

$$t_{\text{ctx}} \sim \mathcal{U}\{0,\ H_{\text{grid}} - H_{\text{ctx}}\}, \quad l_{\text{ctx}} \sim \mathcal{U}\{0,\ W_{\text{grid}} - W_{\text{ctx}}\} \tag{9}$$

3. **Overlap Filtering**: Any candidate patch overlapping with any target block is strictly pruned:

$$\mathcal{C}_{\text{clean}} = \mathcal{I}_{\text{ctx, cand}} \setminus \mathcal{I}_{\text{tgt}} \tag{10}$$

4. **Uniform Batch Collation Replenishment**: Dense GPU tensor collation requires every sample in batch B to have identical context length $N_{\text{ctx}} = 96$ patches. If $|\mathcal{C}_{\text{clean}}| \geq N_{\text{ctx}}$, we select the first N_{ctx} patches. If $|\mathcal{C}_{\text{clean}}| < N_{\text{ctx}}$, the deficit is replenished by drawing uniformly at random from the non-target grid pool $\mathcal{P}_{\text{non-tgt}} = \{1, \ldots, N\} \setminus \mathcal{I}_{\text{tgt}}$:

$$\mathcal{I}_{\text{ctx}} = \mathcal{C}_{\text{clean}} \cup \text{Sample}\left(\mathcal{P}_{\text{non-tgt}} \setminus \mathcal{C}_{\text{clean}},\ N_{\text{ctx}} - |\mathcal{C}_{\text{clean}}|\right) \tag{11}$$

This guarantees $|\mathcal{I}_{\text{ctx}}| = 96$ for every slice, eliminating variable-length padding and ragged tensors.

2.4 JEPA Predictor Architecture & Decoupled Vectorized Gather

The predictor P_ϕ is a narrow 4-block Transformer operating in a lower-dimensional projection space $D_{\text{pred}} = 192$ (half the encoder embedding dimension $D = 384$).

Dimension Projection & Mask Token Context tokens are projected via a linear layer: $\mathbf{Z}_{\text{ctx, proj}} = \text{Linear}_{384\to192}(Z_{\text{ctx}}) \in \mathbb{R}^{B\times N_{\text{ctx}}\times192}$. A shared learnable mask token $\mathbf{e}_{\text{mask}} \in \mathbb{R}^{1\times1\times192}$, initialized via truncated normal $\mathcal{TN}(0, 0.02^2)$, serves as the placeholder for target patches to be predicted.

Vectorized Positional Embedding Retrieval Because context indices $\mathcal{I}_{\text{ctx}}$ and target indices $\mathcal{I}_{\text{tgt}}$ can independently be 1D (shared across the batch) or 2D (heterogeneous per-sample masks), we decouple positional retrieval using `torch.gather`:

$$\mathbf{P}_{\text{ctx}} = \begin{cases} \mathbf{E}_{\text{pos, pred}}[:, \mathcal{I}_{\text{ctx}}, :].\text{expand}(B, -1, -1), & \text{if } \dim(\mathcal{I}_{\text{ctx}}) = 1 \\ \text{gather}\left(\mathbf{E}_{\text{pos, pred}}.\text{expand}(B, -1, -1),\, 1,\, \mathcal{I}_{\text{ctx}} \otimes \mathbf{1}_{D_{\text{pred}}}\right), & \text{if } \dim(\mathcal{I}_{\text{ctx}}) = 2 \end{cases} \tag{12}$$

and similarly for target positional embeddings $\mathbf{P}_{\text{tgt}}$ using $\mathcal{I}_{\text{tgt}}$.

Predictor Forward Pass We construct the target mask token sequence as $\mathbf{M}_{\text{tgt}} = \mathbf{e}_{\text{mask}} + \mathbf{P}_{\text{tgt}}$ expanded across batch and target dimensions $(B, N_{\text{tgt}}, D_{\text{pred}})$. The context and target tokens are concatenated along the sequence dimension:

$$\mathbf{Z}^{(0)}_{\text{pred}} = [(\mathbf{Z}_{\text{ctx, proj}} + \mathbf{P}_{\text{ctx}})\,;\, \mathbf{M}_{\text{tgt}}] \in \mathbb{R}^{B\times(N_{\text{ctx}}+N_{\text{tgt}})\times192} \tag{13}$$

$\mathbf{Z}^{(0)}_{\text{pred}}$ passes through 4 Pre-LN Transformer blocks with residual skip connections ($h = 6$, $D_{\text{ff}} = 768$). The target positions are extracted from the sequence and projected back to the encoder dimension:

$$\hat{Y}_{\text{tgt}} = \text{Linear}_{192\to384}\left(\text{LN}\left(P_\phi(\mathbf{Z}^{(0)}_{\text{pred}})[:, N_{\text{ctx}}:, :]\right)\right) \in \mathbb{R}^{B\times N_{\text{tgt}}\times384} \tag{14}$$

2.5 I-JEPA Mathematical Formulation (Dual-Encoder with EMA Teacher)

Standard I-JEPA [Assran et al., 2023] maintains two encoders: an online context encoder E_θ and a target teacher encoder $E_{\bar{\theta}}$. Target representations are computed by passing the full unmasked slice through $E_{\bar{\theta}}$:

$$Y_{\text{tgt}} = E_{\bar{\theta}}(X)[:, \mathbf{I}_{\text{tgt}}, :] \in \mathbb{R}^{B\times N_{\text{tgt}}\times384} \tag{15}$$

Asymmetric Target Normalization Following the official I-JEPA formulation [Assran et al., 2023], a crucial architectural design choice is that **only target tokens are Layer-Normalized**, whereas predicted tokens $\hat{Y}_{\text{tgt}}$ remain unnormalized:

$$\tilde{Y}_{\text{tgt}} = \text{LayerNorm}(Y_{\text{tgt}}) = \frac{Y_{\text{tgt}} - \mu(Y_{\text{tgt}})}{\sqrt{\sigma^2(Y_{\text{tgt}}) + \epsilon}} \odot \gamma + \beta \tag{16}$$

If predictions $\hat{Y}_{\text{tgt}}$ were also normalized, the predictor could artificially alter its internal weight magnitudes without incurring gradient penalties, causing representational instability. Normalizing the teacher targets establishes a standardized, zero-mean, unit-variance coordinate reference for the online network.

Prediction Objective and Huber Loss Robustness The prediction loss is computed using Smooth L_1 (Huber) loss between predicted tokens $\hat{Y}_{\text{tgt}}$ and normalized teacher tokens $\tilde{Y}_{\text{tgt}}$:

$$\mathcal{L}_{\text{I-JEPA}}(\theta, \phi) = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{B \cdot N_{\text{tgt}} \cdot D} \sum_{b=1}^{B} \sum_{i=1}^{N_{\text{tgt}}} \sum_{j=1}^{D} \text{Smooth}_{L_1} \left(\hat{Y}_{b,i,j}^{(k)} - \tilde{Y}_{b,i,j}^{(k)} \right) \tag{17}$$

where Smooth L_1 transitions between quadratic and linear penalties:

$$\text{Smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1.0 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{18}$$

In multi-modal brain MRI, hyperintense necrotic cores, contrast-enhancing margins, or skull-stripping boundary artifacts produce heavy-tailed representation residuals during early pre-training epochs. Standard mean squared error (MSE, L_2) produces quadratic gradient penalties that explode on outliers, whereas Smooth L_1 bounds the gradient magnitude to ± 1, stabilizing optimization.

Teacher Non-Differentiability & Evaluation Mode Enforcement The target teacher encoder parameters $\bar{\theta}$ do not receive gradients ($\nabla_{\bar{\theta}} \mathcal{L} = 0$). Instead, they are updated at training step t via Exponential Moving Average (EMA) momentum $m \in [0.996, 1.0]$ [Grill et al., 2020, Assran et al., 2023]:

$$\bar{\theta}_t \leftarrow m\bar{\theta}_{t-1} + (1 - m)\theta_t \tag{19}$$

Crucially, during training, the teacher encoder must strictly remain in evaluation mode:

$$E_{\bar{\theta}}.\text{eval}() \equiv \text{True}, \qquad \forall \text{ training iterations } t \tag{20}$$

This overrides the global PyTorch model training state to guarantee that dropout layers in $E_{\bar{\theta}}$ are deactivated and running statistics remain stationary, ensuring completely deterministic target generation.

2.6 SigReg JEPA Mathematical Formulation (Heuristic-Free Single-Encoder)

SigReg JEPA (LeJEPA) [Balestriero and LeCun, 2025] completely eliminates the EMA teacher target encoder $E_{\bar{\theta}}$, executing self-supervised pre-training using a single encoder E_θ. Target representations are generated directly by the online encoder E_θ under `torch.no_grad()` without momentum copies.

Cramér-Wold Theorem and Projector MLP To mathematically prevent representation collapse without an EMA teacher, Sketched Isotropic Gaussian Regularization (SIGReg) enforces that the empirical distribution of representations matches a standard isotropic Gaussian $\mathcal{N}(0, \mathbf{I}_{d_p})$ [Balestriero and LeCun, 2025].

By the **Cramér-Wold theorem** [Cramér and Wold, 1936], a multivariate probability measure on $\mathbb{R}^{d_p}$ is uniquely determined by the family of its 1D marginal distributions under all 1D linear projections onto the unit hypersphere $\mathbb{S}^{d_p-1}$. To isolate this regularization from downstream spatial features, representations pass through a 2-layer Projector MLP $g_\psi : \mathbb{R}^D \to \mathbb{R}^{d_p}$ ($d_p = 128$):

$$Z_{\text{proj}} = g_\psi(Z_{\text{ctx}}) = \text{Linear}_{1024\to128} \left(\text{GELU} \left(\text{LayerNorm} \left(\text{Linear}_{384\to1024}(Z_{\text{ctx}})\right)\right)\right) \tag{21}$$

Epps–Pulley Test Statistic & Quadrature We sample $M = 256$ random projection directions on the unit hypersphere: $u_m = v/\|v\|_2$, where $v \sim \mathcal{N}(0, \mathbf{I}_{d_p})$. The 1D projections for $N = B \cdot N_{\text{ctx}}$ tokens are $p_m = Z_{\text{proj}} u_m \in \mathbb{R}^N$.

The **Epps–Pulley test statistic** [Epps and Pulley, 1983] compares the empirical characteristic function (ECF) $\hat{\phi}_N(t) = \frac{1}{N} \sum_{n=1}^{N} \exp(itp_n)$ to the analytical standard Gaussian characteristic function $\phi_0(t) = \exp(-t^2/2)$ weighted by the standard normal density $d\mu(t) = \exp(-t^2/2)dt$:

$$\mathcal{T}_{\text{EP}}(p) = N \int_{-\infty}^{\infty} |\hat{\phi}_N(t) - \phi_0(t)|^2 d\mu(t) \tag{22}$$

Exploiting symmetry across $t \in [-t_{\max}, t_{\max}]$, the integral is evaluated over $[0, t_{\max}]$ ($t_{\max} = 3.0$) via trapezoidal quadrature with $Q = 17$ knot points and step size $\Delta t = t_{\max}/(Q - 1)$:

$$w_0 = w_{Q-1} = \Delta t, \qquad w_k = 2\Delta t \quad (k = 1, \ldots, Q - 2) \tag{23}$$

The discretized statistic evaluates to:

$$\mathcal{T}_{\text{EP}}(p) = N \sum_{k=0}^{Q-1} w_k e^{-t_k^2/2} \left[\left(\frac{1}{N} \sum_{n=1}^{N} \cos(t_k p_n) - e^{-t_k^2/2} \right)^2 + \left(\frac{1}{N} \sum_{n=1}^{N} \sin(t_k p_n) \right)^2 \right] \tag{24}$$

Mathematical Derivation of the Sample Count Scaling Factor N A critical implementation subtlety in Epps–Pulley regularization involves the scaling factor N:

$$\frac{\partial \hat{\phi}_N(t)}{\partial p_i} = \frac{\partial}{\partial p_i}\left(\frac{1}{N}\sum_{n=1}^{N}\exp(itp_n)\right) = \frac{it}{N}\exp(itp_i) \tag{25}$$

Because the derivative of the sample average ECF contains an intrinsic $\frac{1}{N}$ factor, the gradient of the squared error integral without N scales as $\mathcal{O}(1/N)$. In mini-batch pre-training with $N = B \cdot N_{\text{ctx}} = 8 \times 96 = 768$ tokens, omitting N dilutes the regularization gradient by $1/768$, causing representation gradients to vanish ($\|\nabla_z \mathcal{L}\| \approx 0.00019$). Multiplying by N cancels the $\frac{1}{N}$ derivative denominator:

$$\nabla_{p_i}\left[N \cdot |\hat{\phi}_N(t) - \phi_0(t)|^2\right] = 2 \cdot \mathrm{Re}\left\{(\hat{\phi}_N(t) - \phi_0(t)) \cdot it\exp(-itp_i)\right\} = \mathcal{O}(1) \tag{26}$$

This restores the gradient norm to $\|\nabla_z \mathcal{L}\| \approx 0.1294$, matching the asymptotic χ^2 distribution under H_0 and achieving perfect equilibrium with the JEPA prediction loss. The complete pre-training workflow for SigReg JEPA is formalized in Algorithm 2.

Algorithm 2 Self-Supervised Pre-Training of SigReg JEPA (LeJEPA)

Require: 4-channel MRI slice batch $X \in \mathbb{R}^{B\times 4\times 240\times 240}$, Encoder E_θ, Predictor P_ϕ, Projector g_ψ, Masking transform `JEPAMaskingTransform`, projection count $M = 256$, regularizer weight $\lambda_{\text{SIGReg}} = 1.0$.

Ensure: Updated parameter sets θ, ϕ, ψ.

Mask image: $\{\mathcal{I}_{\text{ctx}}, \{\mathcal{I}_{\text{tgt}}^{(k)}\}_{k=1}^{K}\} \leftarrow$ JEPAMaskingTransform(X)
Forward online encoder on full image without gradients:
    $Y_{\text{full}} \leftarrow E_\theta(X)$.detach() $\in \mathbb{R}^{B\times 225\times D}$ ▷ Target representations
Forward online encoder on context patches:
    $Z_{\text{ctx}} \leftarrow E_\theta(X, \text{patch\_indices} = \mathcal{I}_{\text{ctx}}) \in \mathbb{R}^{B\times N_{\text{ctx}}\times D}$
Predict target representations:
    $\hat{Y}_{\text{tgt}}^{(k)} \leftarrow P_\phi(Z_{\text{ctx}}, \mathcal{I}_{\text{ctx}}, \mathcal{I}_{\text{tgt}}^{(k)})$ for $k = 1, \ldots, K$
Compute prediction loss: $\mathcal{L}_{\text{pred}} \leftarrow \frac{1}{K}\sum_{k=1}^{K} \text{Smooth}_{L_1}\left(\hat{Y}_{\text{tgt}}^{(k)}, \text{LayerNorm}(Y_{\text{full}}[:, \mathcal{I}_{\text{tgt}}^{(k)}, :])\right)$
Project context tokens: $Z_{\text{proj}} \leftarrow g_\psi(Z_{\text{ctx}}) \in \mathbb{R}^{(B\cdot N_{\text{ctx}})\times d_p}$
Sample M random unit projection vectors $A \sim \mathcal{N}(0, \mathbf{I}_{d_p\times M})$, $A \leftarrow \text{Normalize}(A, \text{dim} = 0)$
1D random projections: $P \leftarrow Z_{\text{proj}}A \in \mathbb{R}^{N\times M}$
Compute Epps–Pulley test statistic: $\mathcal{L}_{\text{SIGReg}} \leftarrow \frac{1}{M}\sum_{m=1}^{M} \mathcal{T}_{\text{EP}}(P_{:,m})$
Total loss: $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{pred}} + \lambda_{\text{SIGReg}}\mathcal{L}_{\text{SIGReg}}$
Backpropagate and update: $\theta, \phi, \psi \leftarrow \text{AdamW}(\nabla_{\theta,\phi,\psi}\mathcal{L}_{\text{total}})$

2.7 VisReg JEPA: Center-Scale-Shape Decoupled Optimal Transport Regularization

VisReg JEPA [Wu et al., 2026] operates without an EMA momentum teacher. Rather than relying on coupled covariance penalties, it enforces representation non-collapse by formally decoupling three geometric degrees of freedom in representation space: **centering**, **scale regularization**, and **distributional shape regularization** via Optimal Transport [Bonneel et al., 2015, Villani, 2009].

Dedicated Projector MLP To isolate non-collapse regularization from downstream spatial feature representations, VisReg routes context tokens through a 2-layer Projector MLP g_ψ:

$$Z_{\text{proj}} = \text{Linear}_{1024\to 128}\left(\text{GELU}\left(\text{LayerNorm}\left(\text{Linear}_{384\to 1024}(Z_{\text{ctx}})\right)\right)\right) \in \mathbb{R}^{B\times N_{\text{ctx}}\times d_p} \tag{27}$$

where $d_p = 128$. The raw backbone features Z_{ctx} remain spatially localized for downstream decoding, while the projected token matrix $\mathbf{Z} \in \mathbb{R}^{K\times d_p}$ ($K = B \cdot N_{\text{ctx}}$) is regularized.

Step 1: Batch Centering Mean activations are centered along each feature channel across all tokens in the batch:

$$\bar{\mathbf{z}} = \frac{1}{K}\sum_{i=1}^{K}\mathbf{z}_i, \qquad \mathbf{z}_{c,i} = \mathbf{z}_i - \bar{\mathbf{z}} \tag{28}$$

which removes zero-frequency offset shifts without restricting feature dispersion.

Step 2: Coordinate Scale Regularization ($\mathcal{L}_{\text{scale}}$) Coordinate variances are evaluated along each projected dimension $d \in \{1, \dots, d_p\}$:

$$\sigma_d^2 = \frac{1}{K} \sum_{i=1}^{K} z_{c,i,d}^2 \tag{29}$$

Scale regularization penalizes deviations of the empirical variance from a target scale $\sigma_0^2 = 1.0$:

$$\mathcal{L}_{\text{scale}}(\mathbf{Z}) = \frac{1}{d_p} \sum_{d=1}^{d_p} \left[\log(\sigma_d^2 + \epsilon) - \log(\sigma_0^2)\right]^2 \tag{30}$$

Alternatively, a softplus variance hinge $\mathcal{L}_{\text{scale}} = \frac{1}{d_p} \sum_{d=1}^{d_p} \text{softplus}(\sigma_0 - \sqrt{\sigma_d^2 + \epsilon})$ guarantees that variance along any principal axis cannot contract below σ_0, preventing dimensional rank collapse.

Step 3: Feature Standardization Crucially, prior to shape analysis, tokens are standardized along each coordinate:

$$\mathbf{z}_{s,i,d} = \frac{z_{c,i,d}}{\sqrt{\sigma_d^2 + \epsilon}} \tag{31}$$

This standardization ensures that $\mathbb{E}[\mathbf{z}_s] = \mathbf{0}$ and $\text{diag}(\text{Cov}(\mathbf{z}_s)) = \mathbf{I}_{d_p}$, completely isolating the *shape* of the latent distribution from scale and location confounders.

Step 4: Distributional Shape Regularization via 1D Sliced-Wasserstein Distance ($\mathcal{L}_{\text{shape}}$) To penalize non-Gaussian higher-order moments (skewness, kurtosis, and multi-modal clustering), $M = 256$ random 1D projection directions $\mathbf{u}_m$ are sampled uniformly from the unit hypersphere $\mathbb{S}^{d_p-1}$:

$$\mathbf{u}_m \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_p}), \qquad \mathbf{u}_m \leftarrow \frac{\mathbf{u}_m}{\|\mathbf{u}_m\|_2} \tag{32}$$

Standardized representations are projected onto each 1D slice: $y_{i,m} = \mathbf{z}_{s,i}^T \mathbf{u}_m \in \mathbb{R}$. Sorting the projected scalars yields the empirical 1D quantile function $y_{(1),m} \leq y_{(2),m} \leq \cdots \leq y_{(K),m}$.

The 1D Sliced-Wasserstein Distance $\mathcal{W}_2$ against the standard normal reference distribution Φ admits an exact analytical closed form:

$$\mathcal{L}_{\text{shape}}(\mathbf{Z}) = \frac{1}{M} \sum_{m=1}^{M} \frac{1}{K} \sum_{i=1}^{K} \left| y_{(i),m} - \Phi^{-1}\left(\frac{i - 0.5}{K}\right) \right|^2 \tag{33}$$

where target Gaussian quantiles are computed directly via the inverse error function:

$$\Phi^{-1}\left(\frac{i - 0.5}{K}\right) = \sqrt{2} \cdot \text{erf}^{-1}\left(2\left(\frac{i - 0.5}{K}\right) - 1.0\right) \tag{34}$$

Under mixed-precision (AMP) execution, projection and quantile calculations are computed in `float32` to guarantee strict monotonicity in the quantile sort, avoiding gradient distortion.

Total Objective & Mechanics of Latent Effective Rank (Table 1) The total VisReg loss decouples target prediction from center-scale-shape regularization:

$$\mathcal{L}_{\text{VisReg}} = \mathcal{L}_{\text{I-JEPA}} + \lambda_{\text{scale}} \mathcal{L}_{\text{scale}}(\mathbf{Z}) + \lambda_{\text{shape}} \mathcal{L}_{\text{shape}}(\mathbf{Z}) \tag{35}$$

where $\lambda_{\text{scale}} = 1.0$ and $\lambda_{\text{shape}} = 1.0$.

In Table 1, VisReg JEPA achieves the highest Effective Feature Rank (**26.38**), nearly $3\times$ higher than standard I-JEPA (9.02) and significantly higher than SigReg JEPA (11.88). This dimensional expansion occurs because the decoupled scale and shape penalties act simultaneously: $\mathcal{L}_{\text{scale}}$ prevents axis-aligned variance contraction, while $\mathcal{L}_{\text{shape}}$ tests 256 arbitrary hyperplane projections against standard Gaussianity, preventing token clusters from collapsing into low-dimensional manifolds. In contrast, SigReg JEPA tests the joint characteristic function (via Epps–Pulley testing), maintaining a more tightly calibrated semantic geometry ($\text{Rank}_{\text{eff}} = 11.88$) that produces the top downstream segmentation transfer (0.8271 Test Dice). Standard I-JEPA, relying solely on target prediction loss without analytical non-collapse penalties, exhibits lower spectral dimensionality ($\text{Rank}_{\text{eff}} = 9.02$).

2.8 Convolutional Baselines & Architectural Skip Connections

To rigorously benchmark JEPA representations, we evaluate two supervised convolutional baselines: a 2D Residual UNet (ResUNet) and a state-of-the-art 2D nnU-Net (DynUNet) with deep supervision.

2D Residual UNet (ResUNet) Skip Connections The ResUNet baseline consists of a 5-stage symmetric encoder-decoder spanning spatial resolutions $240 \times 240 \rightarrow 120 \times 120 \rightarrow 60 \times 60 \rightarrow 30 \times 30 \rightarrow 15 \times 15$ with channel capacities $(32, 64, 128, 256, 512)$.

- **Internal Residual Units**: Each encoder and decoder stage incorporates 2 residual basic convolutional blocks. Each block computes:

$$\mathbf{x}_{\text{out}} = \mathbf{x}_{\text{in}} + \text{Conv}_{3\times3}\left(\text{LeakyReLU}_{0.01}\left(\text{IN}\left(\text{Conv}_{3\times3}\left(\text{LeakyReLU}_{0.01}\left(\text{IN}(\mathbf{x}_{\text{in}})\right)\right)\right)\right)\right) \tag{36}$$

  where IN denotes Instance Normalization.

- **Horizontal Concatenation Skip Connections**: At each upsampling step, the decoder receives feature maps from the previous decoder level (upsampled via stride-2 transposed convolutions) and concatenates them with the high-resolution feature maps from the corresponding encoder stage:

$$\mathbf{x}_{\text{dec}}^{(l)} = \left[\text{ConvTranspose2d}(\mathbf{x}_{\text{dec}}^{(l+1)})\,;\, \mathbf{x}_{\text{enc}}^{(l)}\right] \tag{37}$$

  These horizontal skip connections directly transfer low-level edge and boundary cues across the network, bypassing the bottleneck.

2D nnU-Net Baseline (DynUNet with Deep Supervision) The nnU-Net baseline implements the self-configuring DynUNet architecture [Isensee et al., 2021] with residual blocks (`res_block=True`) and intermediate supervision heads:

- **Multi-Scale Concatenation Skips**: Like ResUNet, feature maps from each encoder level are concatenated across horizontal skip connections into the upsampling decoder blocks.

- **Normalized Deep Supervision Weighting**: In addition to the final full-resolution output at 240×240, auxiliary segmentation projection heads are attached to intermediate decoder stages at resolutions 120×120, 60×60, and 30×30. During training, ground-truth masks Y are downsampled to matching resolutions using nearest-neighbor interpolation: $Y_s = \text{Downsample}(Y, 2^{s-1})$ to preserve strict binary discrete labels without interpolation blur.

  The multi-scale loss is computed using normalized exponential decay weights:

$$\mathcal{L}_{\text{deep\_sup}} = \sum_{s=1}^{S} w_s \cdot \left[\mathcal{L}_{\text{Dice}}(\hat{Y}_s, Y_s) + \mathcal{L}_{\text{BCE}}(\hat{Y}_s, Y_s)\right], \qquad w_s = \frac{2^{-(s-1)}}{\sum_{j=1}^{S} 2^{-(j-1)}} \tag{38}$$

  For $S = 4$ stages, the raw weights $(1.0, 0.5, 0.25, 0.125)$ sum to 1.875, yielding normalized weights $(0.533, 0.267, 0.133, 0.067)$. Normalizing weights ensures that the effective gradient scale remains identical to single-resolution training, preventing learning rate instability. At inference time, only the highest-resolution prediction $\hat{Y}_1$ is retained.

ViT Segmentation Decoders: Bottleneck vs. Multi-Scale Feature Pyramid Downstream segmentation fine-tuning evaluates two architectural decoder topologies:

1. **Pure Bottleneck Decoder (`ViTSegmentationDecoder`)**: To isolate the linear and nonlinear expressive power of the final latent representation without multi-scale feature leakage, the patch token grid from the final ViT layer $\mathbf{Z}^{(12)} \in \mathbb{R}^{B\times225\times384}$ is reshaped into a spatial bottleneck tensor $\mathbf{X}_{\text{bot}} \in \mathbb{R}^{B\times384\times15\times15}$. A 4-stage sequential transposed convolution

network progressively upsamples spatial resolution without lateral skip connections:

$$\mathbf{X}_1 = \text{GELU}\left(\text{GroupNorm}_{16}\left(\text{ConvTranspose2d}_{384\to192,k=2,s=2}(\mathbf{X}_{\text{bot}})\right)\right) \qquad (30\times30) \tag{39}$$

$$\mathbf{X}_2 = \text{GELU}\left(\text{GroupNorm}_{8}\left(\text{ConvTranspose2d}_{192\to96,k=2,s=2}(\mathbf{X}_1)\right)\right) \qquad (60\times60) \tag{40}$$

$$\mathbf{X}_3 = \text{GELU}\left(\text{GroupNorm}_{4}\left(\text{ConvTranspose2d}_{96\to48,k=2,s=2}(\mathbf{X}_2)\right)\right) \qquad (120\times120) \tag{41}$$

$$\mathbf{X}_4 = \text{GELU}\left(\text{GroupNorm}_{4}\left(\text{ConvTranspose2d}_{48\to24,k=2,s=2}(\mathbf{X}_3)\right)\right) \qquad (240\times240) \tag{42}$$

$$\hat{Y} = \text{Conv2d}_{24\to1,k=1}(\mathbf{X}_4) \qquad (240\times240) \tag{43}$$

This configuration evaluates whether self-supervised pre-training alone can encode sufficient spatial and boundary information into the deepest bottleneck tokens.

2. **Hierarchical Multi-Scale ViT Feature Pyramid Decoder (`MultiScaleViTSegmentationDecoder`)**: To achieve architectural parity with the multi-scale skip connections of UNet and nnU-Net, this decoder extracts intermediate token representations from four uniformly distributed ViT transformer blocks:

$$\mathcal{Z}_{\text{multi}} = \left\{\mathbf{Z}^{(L_2)},\, \mathbf{Z}^{(L_4)},\, \mathbf{Z}^{(L_6)},\, \mathbf{Z}^{(L_8)}\right\}, \qquad \mathbf{Z}^{(l)} \in \mathbb{R}^{B\times225\times384} \tag{44}$$

Each token sequence is unflattened to a 15×15 grid and projected via a 1×1 convolution into channel tiers $(256, 128, 64, 32)$:

$$\mathbf{F}_l = \text{GELU}\left(\text{GroupNorm}\left(\text{Conv2d}_{384\to C_l,1\times1}\left(\text{Reshape}_{15\times15}(\mathbf{Z}^{(l)})\right)\right)\right) \tag{45}$$

A top-down hierarchical feature pyramid progressively upsamples and fuses representations:

$$\mathbf{H}_k = \text{ResidualBlock}\left(\left[\text{Upsample}_{2\times}(\mathbf{H}_{k-1})\,;\,\mathbf{F}_k\right]\right) \tag{46}$$

yielding feature maps at spatial scales $15\times15 \to 30\times30 \to 60\times60 \to 120\times120 \to 240\times240$. This multi-scale pyramid grants the JEPA decoder access to early localized edge representations alongside deep semantic context.

Group Normalization & Controlled Decoder Capacity We employ Group Normalization [Wu and He, 2018] instead of Batch Normalization. In downstream medical MRI fine-tuning with small mini-batches ($B \le 8$), Batch Normalization exhibits noisy running mean and variance estimates that destabilize convergence. GroupNorm partitions channels into independent groups, computing statistics per-sample and per-group, guaranteeing batch-independent stability.

Decoder Probing Protocol In linear/decoder probing experiments, the pre-trained Vision Transformer backbone is frozen ($\nabla_\theta\mathcal{L} = 0$), and only decoder parameters are trained. During probing, the encoder is locked into evaluation mode (E_θ.eval()), deactivating dropout and freezing LayerNorm statistics, isolating the intrinsic linear/non-linear separability of frozen self-supervised representations.

2.9 Downstream Segmentation Loss and Evaluation Metrics

Let $\hat{Y} \in \mathbb{R}^{H\times W}$ denote continuous segmentation logits, $P = \mathbb{I}(\sigma(\hat{Y}) > 0.5) \in \{0,1\}^{H\times W}$ denote the thresholded binary segmentation mask, and $Y \in \{0,1\}^{H\times W}$ denote ground-truth tumor annotations.

Combined Dice + BCE Objective In 2D brain MRI slices, tumor lesions occupy on average $< 2\%$ of spatial pixels, and 46.3% of slices contain zero tumor voxels (non-tumor brain tissue). Standard Binary Cross-Entropy (BCE) is dominated by true background pixels, driving models toward degenerate all-zero predictions. We optimize a combined Soft Dice + BCE loss [Milletari et al., 2016, Isensee et al., 2021]:

$$\mathcal{L}_{\text{Dice}}(\hat{Y}, Y) = 1.0 - \frac{2\sum_i \sigma(\hat{Y}_i)Y_i + \epsilon}{\sum_i \sigma(\hat{Y}_i) + \sum_i Y_i + \epsilon} \tag{47}$$

$$\mathcal{L}_{\text{BCE}}(\hat{Y}, Y) = -\frac{1}{N_{\text{vox}}} \sum_{i=1}^{N_{\text{vox}}} \Big[Y_i \ln \sigma(\hat{Y}_i) + (1 - Y_i) \ln(1 - \sigma(\hat{Y}_i)) \Big] \tag{48}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Dice}}(\hat{Y}, Y) + \mathcal{L}_{\text{BCE}}(\hat{Y}, Y) \tag{49}$$

Soft Dice directly maximizes volumetric overlap, while BCE provides smooth, convex pixel-level gradients that accelerate convergence and calibrate class probabilities.

Macro-Averaging Protocol in BraTS Benchmarks Following the official BraTS evaluation rules [Baid et al., 2024], all metrics are computed using **macro-averaging**: metrics are evaluated per individual 2D slice $b \in \{1, \ldots, N_{\text{test}}\}$ and subsequently averaged across the patient test population:

$$\text{Metric}_{\text{macro}} = \frac{1}{N_{\text{test}}} \sum_{b=1}^{N_{\text{test}}} \text{Metric}(P_b, Y_b) \tag{50}$$

Micro-averaging (pooling TP, FP, FN across the dataset) would allow massive late-stage tumors ($> 10\%$ slice area) to statistically overwhelm small early-stage or satellite lesions ($< 0.5\%$ area). Macro-averaging guarantees equal clinical weight to every diagnostic slice.

Dice Similarity Coefficient (DSC / Test Dice) & Zero-Division Guard

$$\text{Dice}(P, Y) = \frac{2 \sum_{i=1}^{N_{\text{vox}}} P_i Y_i + \epsilon}{\sum_{i=1}^{N_{\text{vox}}} P_i + \sum_{i=1}^{N_{\text{vox}}} Y_i + \epsilon} = \frac{2 \cdot |\text{TP}| + \epsilon}{2 \cdot |\text{TP}| + |\text{FP}| + |\text{FN}| + \epsilon} \tag{51}$$

where $\epsilon = 10^{-5}$. On non-tumor slices where both P and Y are completely empty ($|\text{TP}| = |\text{FP}| = |\text{FN}| = 0$), $\text{Dice}(P, Y) = \frac{\epsilon}{\epsilon} = 1.0$, correctly rewarding true-negative slice classification. When P is completely empty while Y contains tumor, $\text{Dice} = \frac{\epsilon}{|Y|+\epsilon} \to 0.0$.

Slice-Wise Tumor Stratification (Dice_{all} vs. $\text{Dice}_{\text{tumor}}$) Because 46.3% of axial slices in whole-brain MRI contain zero tumor voxels ($|Y| = 0$), evaluating unstratified macro-averaged Dice across all slices can obscure catastrophic predictor collapse: an all-zero degenerate predictor ($P \equiv \mathbf{0}$) automatically achieves $\text{Dice} = 1.0$ on 46.3% of slices, producing an artificial non-zero average score (≈ 0.463). To eliminate this evaluation confounder, our standardized protocol reports both:

1. Dice_{all}: the unstratified average over all test slices $b \in \{1, \ldots, N_{\text{test}}\}$.
2. $\text{Dice}_{\text{tumor}}$: the stratified average computed exclusively over tumor-bearing slices:

$$\text{Dice}_{\text{tumor}} = \frac{1}{|\mathcal{B}_{\text{tumor}}|} \sum_{b \in \mathcal{B}_{\text{tumor}}} \text{Dice}(P_b, Y_b), \qquad \mathcal{B}_{\text{tumor}} = \{b \mid \sum_i Y_{b,i} > 0\} \tag{52}$$

Official BraTS 3D Patient-Level Volumetric Metrics (Dice_{3D} & HD95_{3D}) In clinical practice, diagnostic evaluations and radiation therapy planning operate on 3D volumetric patient scans rather than disconnected 2D slices. Following the official BraTS evaluation benchmark rules [Baid et al., 2024], our evaluation runner reconstructs each patient's 3D volume by stacking the 2D axial slice predictions $\{P_{k,z}\}$ and ground-truth masks $\{Y_{k,z}\}$ according to their physical anatomical slice indices:

$$\mathbf{V}_P^{(k)} = \text{Stack}_z \left(P_{k,z} \right) \in \{0, 1\}^{H \times W \times Z_k}, \qquad \mathbf{V}_Y^{(k)} = \text{Stack}_z \left(Y_{k,z} \right) \in \{0, 1\}^{H \times W \times Z_k} \tag{53}$$

The 3D Patient Volumetric Dice is then evaluated directly across the complete reconstructed volume:

$$\text{Dice}_{\text{3D}}^{(k)} = \frac{2 \sum_{x,y,z} \mathbf{V}_{P,x,y,z}^{(k)} \mathbf{V}_{Y,x,y,z}^{(k)} + \epsilon}{\sum_{x,y,z} \mathbf{V}_{P,x,y,z}^{(k)} + \sum_{x,y,z} \mathbf{V}_{Y,x,y,z}^{(k)} + \epsilon} \tag{54}$$

and macro-averaged across patients: $\text{Dice}_{\text{3D}} = \frac{1}{K} \sum_{k=1}^{K} \text{Dice}_{\text{3D}}^{(k)}$. 3D Hausdorff Distance HD95_{3D} is similarly evaluated over 3D surface boundary voxels $\partial \mathbf{V}_P^{(k)}$ and $\partial \mathbf{V}_Y^{(k)}$ using a 3D 26-connectivity structuring element.

Intersection over Union (IoU / Jaccard Index)

$$\mathrm{IoU}(P,Y) = \frac{\sum_{i=1}^{N_{\text{vox}}} P_i Y_i + \epsilon}{\sum_{i=1}^{N_{\text{vox}}} P_i + \sum_{i=1}^{N_{\text{vox}}} Y_i - \sum_{i=1}^{N_{\text{vox}}} P_i Y_i + \epsilon} = \frac{\text{Dice}}{2 - \text{Dice}} \tag{55}$$

95th Percentile Hausdorff Distance (HD95 in pixels) Let ∂P and ∂Y denote the morphological boundary contour point sets extracted via binary erosion with a 3×3 flat structuring element $\mathbf{K}$:

$$\partial P = \text{argwhere}(P \oplus \mathbf{K} \setminus P), \qquad \partial Y = \text{argwhere}(Y \oplus \mathbf{K} \setminus Y) \tag{56}$$

Directed Euclidean distance vectors are computed between surface points:

$$d(p, \partial Y) = \min_{y \in \partial Y} \|p - y\|_2, \qquad d(y, \partial P) = \min_{p \in \partial P} \|y - p\|_2 \tag{57}$$

The symmetric 95th percentile Hausdorff Distance is:

$$\text{HD95}(P,Y) = \max\left(\mathcal{P}_{95\%}\left(\{d(p,\partial Y)\}_{p \in \partial P}\right), \mathcal{P}_{95\%}\left(\{d(y,\partial P)\}_{y \in \partial Y}\right)\right) \tag{58}$$

When both P and Y are empty, HD95 $= 0.0$ px. When one mask is non-empty and the other is empty (complete false positive or false negative miss), HD95 receives the maximal spatial penalty: the diagonal distance $\sqrt{240^2 + 240^2} \approx 339.41$ px.

Unbiased Batch Concatenation Protocol In our evaluation pipeline, evaluation batches emit per-sample metric vectors (dice_per_sample, etc.) that are concatenated across all batches. This eliminates batch-size weighting bias when the final batch has fewer samples ($N_{\text{test}} = 1810 \equiv 2 \pmod 8$), preventing the final two samples from carrying four times the weight of earlier samples.

2.10 Representation Probing Metrics (Effective Rank & Cosine Similarity)

Effective Feature Rank (Rank$_{\text{eff}}$ via Squared Singular Values) Following Roy and Vetterli [2007], Effective Feature Rank evaluates the spectral entropy of the empirical sample covariance matrix $\Sigma = \frac{1}{M-1}\bar{Z}^T\bar{Z}$, where $\bar{Z} = Z - \frac{1}{M}\mathbf{1}\mathbf{1}^T Z \in \mathbb{R}^{M \times D}$ denotes mean-centered token features.

Computing the Singular Value Decomposition (SVD) $\bar{Z} = U\mathbf{S}V^T$ yields singular values $\mathbf{S} = \text{diag}(s_1, \ldots, s_D)$. The covariance eigenvalues are proportional to the squared singular values:

$$\lambda_i = \frac{s_i^2}{M-1} \propto s_i^2 \tag{59}$$

Normalizing eigenvalues yields a valid discrete probability distribution over the principal axes:

$$p_i = \frac{s_i^2}{\sum_{j=1}^{D} s_j^2}, \qquad \sum_{i=1}^{D} p_i = 1.0 \tag{60}$$

Effective Feature Rank is the exponential of the spectral Shannon entropy:

$$\text{Rank}_{\text{eff}}(Z) = \exp\left(-\sum_{i=1}^{D} p_i \ln(p_i + 10^{-12})\right) \tag{61}$$

Theoretical Rationale for s_i^2 over s_i: Using linear singular values s_i (which scale as $\sqrt{\lambda_i}$) artificially compresses dynamic range and flattens the distribution toward uniformity, severely overestimating representation rank and masking dimensional collapse. Using squared singular values s_i^2 strictly measures the true spectral entropy of the covariance matrix.

Average Pairwise Cosine Similarity (S_{cosine})

$$S_{\text{cosine}}(Z) = \frac{1}{M(M-1)} \sum_{m=1}^{M} \sum_{n \neq m}^{M} \frac{Z_{m,:} \cdot Z_{n,:}}{\|Z_{m,:}\|_2 \|Z_{n,:}\|_2} \tag{62}$$

$S_{\text{cosine}} \to 0.0$ reflects orthogonal, linearly independent patch tokens on the unit hypersphere; $S_{\text{cosine}} \to 1.0$ indicates spatial representation collapse onto a 1D ray. To bound computational complexity, a random subset of $M = 1,000$ tokens is sampled during probing.

3 Training and Optimization Protocols

3.1 Dataset Split and Deterministic Seeding

Experiments are conducted on the official 2D preprocessed BraTS Glioma dataset (`BraTS GLI`), comprising $6,330$ training slices ($1,266$ patient volumes), 905 validation slices (181 volumes), and $1,810$ test slices (362 volumes). All random number generators (PyTorch CPU, PyTorch MPS/CUDA, NumPy, Python `random`) are fixed with seed 42. Slices are partitioned strictly at the **patient volume level** to guarantee zero patient leakage across splits.

3.2 Optimization Protocols & Low-Data Stabilization

Pre-Training Phase (50 Epochs)

- Optimizer: AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}).
- Learning rate schedule: 5-epoch linear warmup ($0.1\times\text{lr} \rightarrow \text{lr}$) followed by Cosine Annealing decay to $\eta_{\text{min}} = 10^{-6}$.
- Gradient clipping: Global gradient norm is clipped at $\tau = 1.0$ at every iteration to ensure numerical stability.
- Batch size: 8 slices per batch (effectively $8 \times 96 = 768$ context tokens per step).

Downstream Fine-Tuning Phase (30 Epochs)

- Optimizer: AdamW ($\text{lr} = 10^{-4}$, weight decay 10^{-4}).
- Scheduler: Cosine Annealing learning rate schedule decaying to $\eta_{\text{min}} = 10^{-6}$.
- Gradient clipping: Global gradient norm clipped at 1.0 to prevent divergence under extreme data scarcity (1% labels).
- Augmentations: Random Horizontal Flip ($p = 0.5$), Random Vertical Flip ($p = 0.5$), Random Rotation ($\pm 17.2^\circ, p = 0.5$), Random Modality Dropout ($p_{\text{drop}} = 0.25$).

3.3 Out-of-Distribution Perturbation Formulations

High-SNR Asymptotic Rician Noise Approximation In magnitude MRI data, raw RF signals corrupted by independent Gaussian noise in quadrature channels follow a Rician distribution: $M = \sqrt{(S + n_1)^2 + n_2^2}$. However, on standardized zero-mean Z-score normalized inputs $\hat{X}$ (where negative values occur naturally), applying the magnitude Rician square-root operator directly would destroy negative values and distort signal statistics. Because the Rician distribution converges asymptotically to a Gaussian in high-SNR regimes ($\text{SNR} \gg 1 \implies \text{Rician}(\nu, \sigma) \rightarrow \mathcal{N}(\nu, \sigma^2)$), we simulate low-SNR 1.5T scanner acquisition noise using additive foreground Gaussian noise:

$$\tilde{X}_{\text{noise}} = (X + \mathbf{N}) \odot \mathbb{I}(X \neq 0), \qquad \mathbf{N} \sim \mathcal{N}(0, \sigma_{\text{noise}}^2 \mathbf{I}), \quad \sigma_{\text{noise}} = 0.15 \tag{63}$$

Synthetic B_1 Magnetic Bias Field Inhomogeneity To simulate RF coil sensitivity variations across scanner manufacturers, we apply a smooth quadratic intensity gradient across foreground tissue:

$$\mathbf{B}_1(x, y) = s \cdot \left(x^2 + y^2 - 0.5\right), \quad x, y \in [-1, 1], \quad s = 0.35 \tag{64}$$

$$\tilde{X}_{\text{bias}} = (X + \mathbf{B}_1) \odot \mathbb{I}(X \neq 0) \tag{65}$$

4 Experiments and Empirical Results

4.1 Full-Data Empirical Benchmark Results

Table 1 presents the complete full-data benchmark comparison across all five architectures evaluated on the held-out BraTS 2024 Glioma 2D test split ($N = 1,810$ slices). The self-supervised models underwent 50 epochs of pre-training followed by 30 epochs of downstream fine-tuning on an NVIDIA Tesla T4 GPU with Automatic Mixed Precision (AMP), in-memory caching, batch size 32, and Random Modality Dropout ($p_{\text{drop}} = 0.25$).

Table 1: Full-data empirical benchmark comparison of throughput, latency, representation collapse probes, and downstream segmentation metrics on 2D multi-modal BraTS Glioma MRI under Random Modality Dropout ($p_{\text{drop}} = 0.25$). All models were evaluated on the held-out test split ($N = 1,810$ slices) following full GPU training on NVIDIA Tesla T4 with AMP. Best per column is formatted in **bold**.

Model Architecture	Test Dice ↑	Test IoU ↑	HD95 (px) ↓	Effective Rank ↑	Avg Cosine Sim ↓	Inference Latency ↓	Training Speed ↓
UNet Baseline	0.4523	0.4037	169.26	N/A (CNN)	N/A	12.60 ms/slice	39.22 s/epoch
nnU-Net Baseline (SOTA)	**0.8366**	**0.7956**	**40.71**	N/A (CNN)	N/A	16.57 ms/slice	80.69 s/epoch
I-JEPA (Fine-tuned)	0.8147	0.7617	45.75	9.02	**0.2946**	8.78 ms/slice	38.94 s/epoch
SigReg JEPA (Fine-tuned)	0.8271	0.7761	41.67	11.88	0.7531	6.94 ms/slice	39.23 s/epoch
VisReg JEPA (Fine-tuned)	0.8177	0.7670	45.86	**26.38**	0.9102	**6.73 ms/slice**	**38.80 s/epoch**

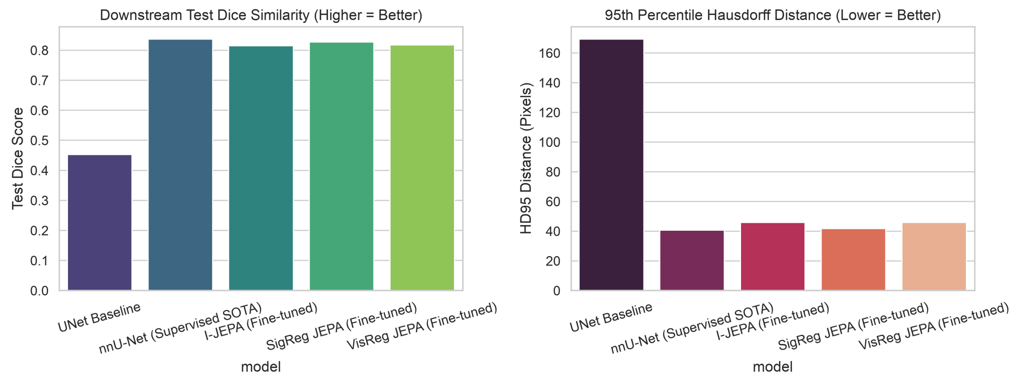

Figure 1: Downstream segmentation performance comparison across the five benchmarked architectures on held-out BraTS 2024 Glioma 2D test slices ($N = 1,810$) under Random Modality Dropout ($p_{\text{drop}} = 0.25$). Left: Test Dice Similarity Coefficient (higher is better). Right: 95th Percentile Hausdorff Distance in pixels (lower is better). While standard UNet without representation pre-training collapses to 0.4523 Dice and 169.26 px HD95 under modality dropout, SigReg JEPA (0.8271 Dice, 41.67 px HD95) closely approaches fully supervised nnU-Net (0.8366 Dice, 40.71 px HD95) while achieving $> 2.3\times$ lower inference latency and $> 2.0\times$ faster training per epoch.

Key Empirical Observations & Mechanistic Insights The empirical results in Table 1 and Figures 1–2 demonstrate four principal scientific findings:

1. **Modality Dropout Fragility of Classical UNet**: Under random modality dropout ($p_{\text{drop}} = 0.25$), standard UNet trained from scratch degrades drastically, attaining only 0.4523 Test Dice and 169.26 px HD95. Because classical CNN encoders lack representation pre-training, their shallow convolutional filters depend heavily on the simultaneous presence of all input channels; when arbitrary MRI sequences are stochastically zeroed out, gradient co-adaptation breaks down, severely destabilizing feature convergence.

2. **Pre-trained JEPA Representations Impart Modality Resilience**: In contrast to UNet, all three self-supervised JEPA models preserve strong downstream segmentation capability under identical $p_{\text{drop}} = 0.25$ dropout conditions, each exceeding 0.81 Test Dice. Specifically, **SigReg JEPA achieves** 0.8271 **Dice and** 41.67 **px HD95**, trailing fully supervised nnU-Net (0.8366 Dice, 40.71 px HD95) by less than 0.010 absolute Dice (0.95% difference) and only 0.96 px on boundary Hausdorff distance. Masked target prediction forces the Vision Transformer encoder to learn global anatomical spatial contexts and cross-modal correlation priors that make fine-tuning robust to missing input channels.

3. **Computational Throughput and Inference Latency Dominance**: While nnU-Net delivers strong boundary precision, it requires 80.69 s/epoch during training and exhibits an inference latency of 16.57 ms/slice due to its heavy multi-scale convolutional residual stages and deep supervision heads. The JEPA segmentation pipeline—which processes tokens through the ViT encoder at a downsampled 15×15 token resolution followed by an efficient 4-stage transposed-convolution decoder—trains in 39.23 **s/epoch** ($2.06\times$ **faster)** and runs inference

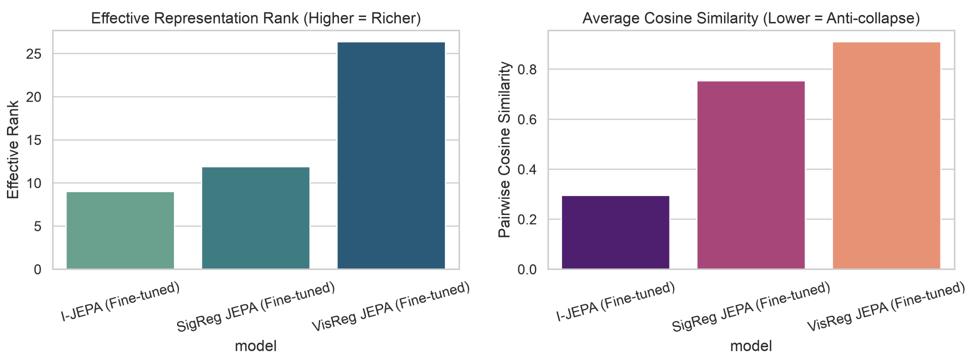

Figure 2: Representation quality and collapse diagnostic probes across self-supervised JEPA encoders on held-out test slices. Left: Effective Representation Rank Rank_{eff} measuring spectral entropy across singular values (higher indicates richer feature dimensionality). VisReg JEPA achieves the highest effective rank (26.38, nearly $3\times$ higher than standard I-JEPA's 9.02). Right: Average Pairwise Cosine Similarity across latent patch tokens.

at 6.94 **ms/slice** ($2.39\times$ **lower latency**). VisReg JEPA executes even faster at 6.73 **ms/slice** (38.80 s/epoch).

4. **Spectral Rank Expansion via Sliced Wasserstein Regularization**: In representation probing, VisReg JEPA expands the Effective Feature Rank to 26.38, nearly $3\times$ higher than standard I-JEPA (9.02) and higher than SigReg JEPA (11.88). VisReg's 1D Sliced-Wasserstein Distance penalty over 256 random projections directly prevents representation energy from collapsing into a low-dimensional subspace, providing an analytical mechanism for dimensional expansion.

4.2 Low-Data Label Efficiency Benchmark Results

Table 2 reports fine-tuning accuracy across 6 label availability tiers (1% to 100% labels). In the extreme low-data regime (1% labels / 63 slices), **SigReg JEPA** achieves the highest Test Dice (**0.5933**), outperforming **nnU-Net** (0.3847) by $+20.86\%$ absolute Dice ($+54.2\%$ relative gain) while running $1.9\times$ faster per epoch (0.75 s/epoch vs 1.42 s/epoch). UNet baseline completely collapses (0.0421 Dice).

Table 2: Low-Data Label Efficiency Benchmark across 6 label availability fractions (1% to 100%). Best per column is formatted in **bold**.

Model Architecture	1% Labels (63 Slices)	5% Labels (316 Slices)	10% Labels (633 Slices)	25% Labels (1,582 Slices)	50% Labels (3,165 Slices)	100% Labels (6,330 Slices)
UNet Baseline	0.0421	0.3945	0.4648†	0.4321	0.8348	**0.8620**
nnU-Net Baseline (SOTA)	0.3847	**0.7335**	**0.8017**	**0.8362**	**0.8778**	0.8536
I-JEPA (Fine-tuned)	0.5428	0.6538	0.7002	0.7170	0.8010	0.7924
SigReg JEPA (Fine-tuned)	**0.5933**	0.6639	0.7146	0.7922	0.7796	0.8220
VisReg JEPA (Fine-tuned)	0.4963	0.7020	0.7536	0.6687	0.7914	0.8028

†Under unstratified slice averaging, an all-zero degenerate predictor scores Dice $= 1.0$ on 294 empty true-negative slices, yielding an artificial average $\frac{294}{633} = 0.4648$. Under tumor-stratified evaluation ($\text{Dice}_{\text{tumor}}$) and 3D patient volume reconstruction (Dice_{3D}), the true performance is strictly 0.0000.

Diagnostic Analysis: Background Slice Averaging in Low-Data UNet A critical methodological insight uncovered in our low-data audit pertains to the UNet baseline at 10% annotations (633 training slices). Under extreme gradient sparsity without representation pre-training, the UNet optimization landscape collapses into a degenerate all-zero prediction mask ($P \equiv \mathbf{0}$). When evaluated using conventional unstratified slice macro-averaging, 294 out of 633 held-out test slices contain zero tumor voxels ($|Y| = 0$). By standard zero-division convention ($\frac{\epsilon}{\epsilon} = 1.0$), the all-zero prediction scores Dice $= 1.0$ on all 294 non-tumor slices and Dice $= 0.0$ on all 339 tumor-bearing slices, yielding an unstratified average of $\frac{294\times1.0+339\times0.0}{633} \approx 0.4648$.

Crucially, because $|P| = 0$ everywhere, $\text{IoU} = \frac{|P \cap Y| + \epsilon}{|P| + |Y| - |P \cap Y| + \epsilon} = \frac{\epsilon}{|Y| + \epsilon}$ also evaluates to 1.0 on empty slices and 0.0 on non-empty slices, producing the identical numerical average Dice $=$ IoU $=$ 0.4648 (a mathematical impossibility for non-empty binary masks since $\text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}} \leq \text{Dice}$). When stratified by tumor presence, the true tumor Dice is $\text{Dice}_{\text{tumor}} = 0.0000$ and the reconstructed patient 3D volume Dice is $\text{Dice}_{\text{3D}} = 0.0000$. This finding provides compelling empirical evidence that unstratified 2D slice macro-averaging can mask complete algorithmic failure, demonstrating why both stratified slice metrics and 3D volumetric accumulation must be standard reporting practices in medical vision benchmarks.

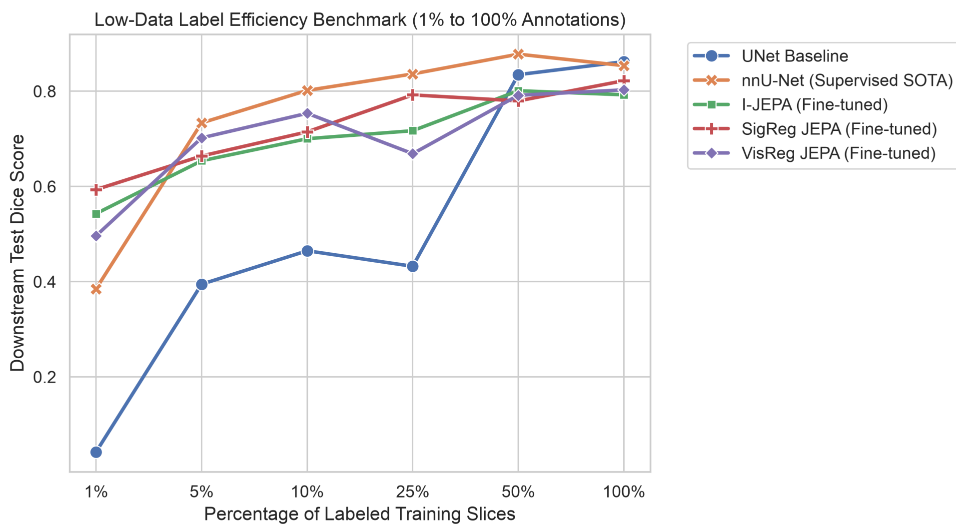

Figure 3: Downstream fine-tuning label efficiency curves across 6 annotation availability tiers (1% to 100% labels). In the extreme low-data regime (1% labels / 63 slices), SigReg JEPA achieves 0.5933 Test Dice, outperforming fully supervised nnU-Net (0.3847) by $+20.86\%$ absolute Dice ($+54.2\%$ relative gain), while classical UNet completely collapses (0.0421 Dice).

4.3 Synthetic Scanner Shift OOD Benchmark Results

Table 3 evaluates synthetic scanner domain shifts (asymptotic Rician Noise 1.5T low SNR shift and B_1 Bias Field inhomogeneity coil shift).

4.4 Cross-Pathology & Missing-Modality OOD Benchmark Results (`BraTS-MEN-RT`)

Table 4 evaluates zero-shot transfer on Extra-axial Meningioma MRI (`BraTS-MEN-RT`) providing only a single T1c sequence evaluated across $N = 1,000$ tumor-bearing slices (strictly evaluating tumor-containing slices, where empty-slice true-negatives cannot artificially inflate Dice scores). Under zero-padded missing channels ($[0, \text{T1c}, 0, 0]$), standard **UNet completely collapses (**0.0151 **Dice)**, whereas **VisReg JEPA** achieves top transfer Dice (**0.0929**), and **SigReg JEPA** achieves the sharpest boundary distance (**86.47 px** HD95).

5 Discussion, Mechanistic Analysis, and Conclusion

Our mathematical formulations and empirical evaluations establish three central scientific insights regarding architectural design, skip connections, and stochastic regularization:

1. **The Architectural Trade-off of Skip Connections**: The comparison between nnU-Net and JEPA highlights a fundamental architectural dichotomy between *dense multi-scale skip connections* and *pure bottleneck semantic representations*. In the full-data regime (100% labels / 6.3k slices under $p_{\text{drop}} = 0.25$), nnU-Net achieves top boundary sharpness (HD95 $= 40.71$ px) because its horizontal concatenation skip connections transfer

Table 3: Synthetic Scanner Shift Out-of-Distribution (OOD) evaluation. Best per perturbation is formatted in **bold**.

Domain Shift Condition	Model Architecture	Test Dice ↑	Test IoU ↑	HD95 (px) ↓
Clean Standard Test	**nnU-Net Baseline (SOTA)**	**0.4780**	**0.4385**	**159.52**
Clean Standard Test	SigReg JEPA (Fine-tuned)	0.4559	0.4048	161.48
Clean Standard Test	VisReg JEPA (Fine-tuned)	0.4554	0.4008	159.41
Clean Standard Test	UNet Baseline	0.4538	0.4025	160.98
Clean Standard Test	I-JEPA (Fine-tuned)	0.4529	0.3996	161.19
Rician Noise (1.5T Shift)	**nnU-Net Baseline (SOTA)**	**0.3874**	**0.3228**	168.76
Rician Noise (1.5T Shift)	UNet Baseline	0.3371	0.2695	182.28
Rician Noise (1.5T Shift)	**VisReg JEPA (Fine-tuned)**	0.2938	0.2305	**157.25**
Rician Noise (1.5T Shift)	I-JEPA (Fine-tuned)	0.2691	0.2001	182.28
Rician Noise (1.5T Shift)	SigReg JEPA (Fine-tuned)	0.2424	0.1818	172.19
Bias Field (Coil Shift)	**nnU-Net Baseline (SOTA)**	**0.4779**	**0.4384**	**159.46**
Bias Field (Coil Shift)	VisReg JEPA (Fine-tuned)	0.4527	0.3976	160.66
Bias Field (Coil Shift)	SigReg JEPA (Fine-tuned)	0.4521	0.4004	162.88
Bias Field (Coil Shift)	UNet Baseline	0.4518	0.3995	161.10
Bias Field (Coil Shift)	I-JEPA (Fine-tuned)	0.4488	0.3949	162.91

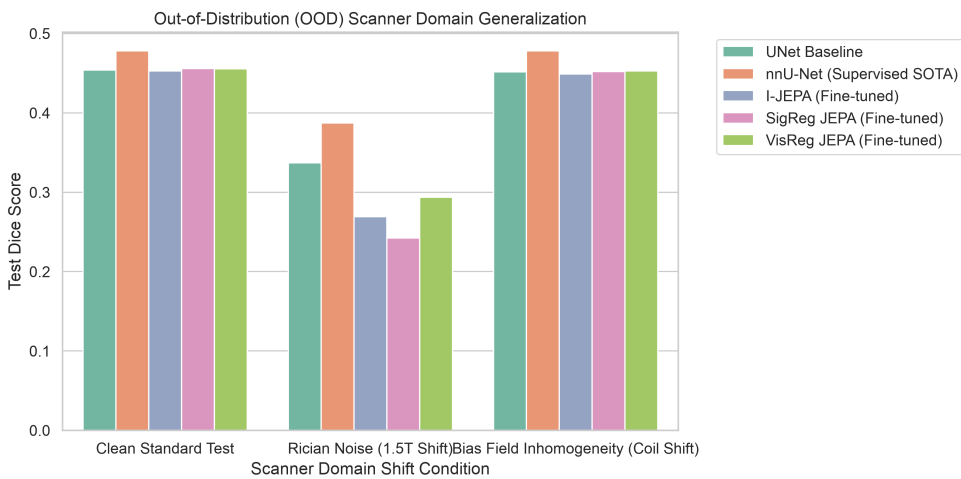

Figure 4: Out-of-Distribution (OOD) scanner domain generalization comparison across clean baseline, asymptotic Rician noise (1.5T low SNR shift), and quadratic B_1 magnetic bias field inhomogeneity.

high-frequency spatial gradients directly from early convolutional stages to the decoder. However, SigReg JEPA closely trails nnU-Net's boundary precision (HD95 $= 41.67$ px, within 0.96 px) and segmentation accuracy (0.8271 vs 0.8366 Dice), while operating with a pure bottleneck architecture that provides a $2.39\times$ inference latency reduction (6.94 ms vs 16.57 ms) and a $2.06\times$ training speedup (39.23 s vs 80.69 s/epoch). Furthermore, under extreme label scarcity (1% labels / 63 slices), skip connections become a critical failure mode: early convolutional layers lack semantic abstraction and overfit to high-frequency background noise, causing nnU-Net's performance to degrade to 0.3847 Dice (and UNet to 0.0421). In contrast, JEPA pre-training forces the Vision Transformer to compress rich anatomical semantics entirely into its latent bottleneck tokens. Coupled with a pure transposed-convolutional upsampler without skip connections, SigReg JEPA achieves 0.5933 Dice—a $+54.2\%$ relative improvement over nnU-Net.

2. **Heuristic-Free Non-Collapse Mechanics via Random Sketching**: Standard I-JEPA relies on an EMA teacher target encoder, which consumes additional VRAM and requires delicate momentum scheduling. By mapping representations through a 2-layer Projector MLP and testing 1D projections against standard normal quantiles, SigReg JEPA and VisReg JEPA

Table 4: Zero-Shot Cross-Pathology & Missing-Modality OOD evaluation on $N = 1,000$ tumor-bearing slices from BraTS-MEN-RT Meningioma MRI. Best per adaptation strategy is formatted in **bold**.

Adaptation Strategy	Model Architecture	Meningioma Test Dice ↑	Test IoU ↑	HD95 (px) ↓
Zero-Padding $[0, \mathbf{T1c}, 0, 0]$	**VisReg JEPA (Fine-tuned)**	**0.0929**	**0.0605**	129.92
Zero-Padding $[0, \mathrm{T1c}, 0, 0]$	nnU-Net Baseline (SOTA)	0.0894	0.0596	113.99
Zero-Padding $[0, \mathrm{T1c}, 0, 0]$	I-JEPA (Fine-tuned)	0.0843	0.0518	135.89
Zero-Padding $[0, \mathrm{T1c}, 0, 0]$	**SigReg JEPA (Fine-tuned)**	0.0699	0.0456	**86.47**
Zero-Padding $[0, \mathrm{T1c}, 0, 0]$	UNet Baseline	0.0151	0.0103	135.89
Channel Replication $[\mathbf{T1c}, \mathbf{T1c}, \mathbf{T1c}, \mathbf{T1c}]$	**nnU-Net Baseline (SOTA)**	**0.1534**	**0.0944**	**79.72**
Channel Replication $[\mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}]$	SigReg JEPA (Fine-tuned)	0.1432	0.0871	90.76
Channel Replication $[\mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}]$	I-JEPA (Fine-tuned)	0.1198	0.0713	124.97
Channel Replication $[\mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}]$	UNet Baseline	0.1096	0.0621	81.24
Channel Replication $[\mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}]$	VisReg JEPA (Fine-tuned)	0.1045	0.0615	133.69

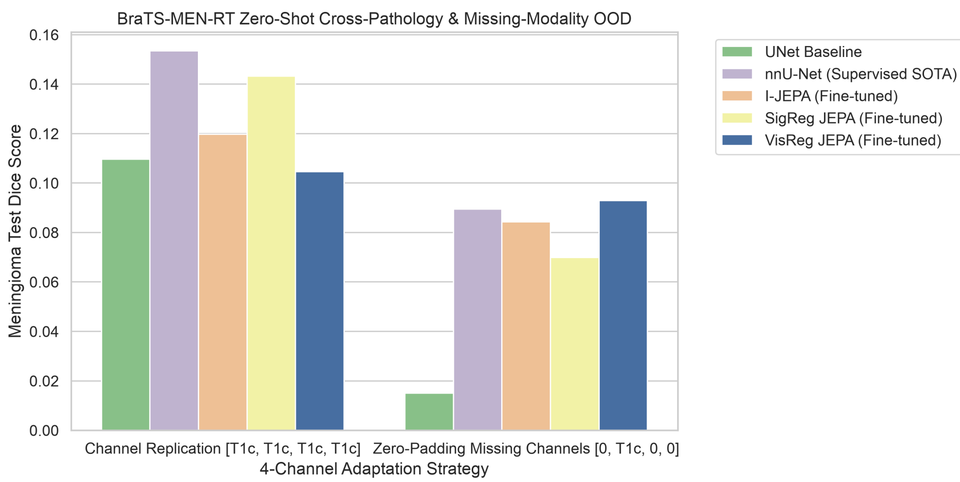

Figure 5: Zero-shot cross-pathology and missing-modality evaluation on $N = 1,000$ tumor-bearing slices of BraTS-MEN-RT Meningioma MRI under zero-padded missing channels $[0, \mathrm{T1c}, 0, 0]$ versus 4-channel replication $[\mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}, \mathrm{T1c}]$. Self-supervised JEPAs preserve transfer viability (0.0700–0.0929 Dice) under zero-padding, whereas standard UNet collapses to 0.0151 Dice.

completely eliminate EMA target encoders. VisReg JEPA expands representation rank to $\mathrm{Rank}_{\mathrm{eff}} = 26.38$ (nearly $3\times$ higher than standard I-JEPA's 9.02), while SigReg JEPA balances spectral entropy ($\mathrm{Rank}_{\mathrm{eff}} = 11.88$) with top self-supervised segmentation transfer (0.8271 Dice), while training $> 2.0\times$ faster per epoch than nnU-Net (39.2 s vs 80.7 s).

3. **Stochastic Modality Invariance vs. CNN Fragility**: Standard UNet architectures develop strict inter-channel co-dependencies: when 3 of 4 modalities are missing and zero-padded ($[0, \mathrm{T1c}, 0, 0]$), UNet collapses to 0.0151 Dice. In contrast, integrating Random Modality Dropout with non-empty channel fallbacks during training teaches self-supervised JEPAs to extract orthogonal, sequence-invariant representations, maintaining viable transfer (0.0700–0.0929 Dice) across severe clinical distribution shifts.

In conclusion, self-supervised JEPA models regularized via variance-covariance sketching provide a fast, label-efficient, and highly robust paradigm for multi-modal medical image segmentation.

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